

Part I.

1. The **Free-Body Diagram (FBD)** of a rigid body represents it as having being isolated from its surroundings, replacing physical supports with forces and couple moments.
2. True or **False**. An ideal truss does not have three-force members
3. For 3D systems, the internal moment has two components: **Shear Force** and **Normal Force**, whose axes are tangent and normal to the cross section, respectively.
4. A body is said to be **Under constrained** if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.
5. True or **False**. A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate.
6. For a simply supported thirteen-member plane truss to be considered simple, it must have **7(seven)** joints.
7. True or **False**. A frictionless ball-and-socket joint resists translations and rotations.
8. A body is considered **improperly** constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.
9. **True** or False. A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.
10. For internal loadings of 2D systems, the **Normal** force and the **Shear** force are normal and tangent to the cross section, respectively.

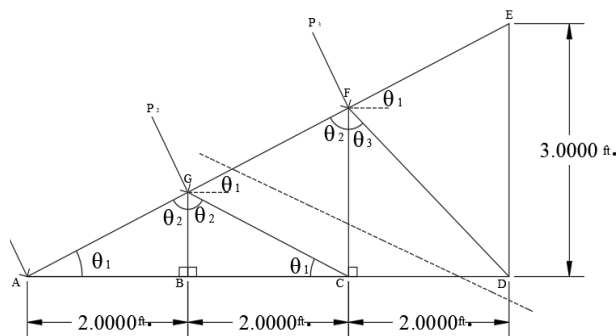
Scenario 1. A 1.1-m long cantilever beam is subject to a 2 kN weight at its free end.

11. **True** or False. This scenario is statically determinate.
12. The support reactions have magnitudes of **2** kN and **2.2** kN-m.
13. The maximum bending moment developed is **2.2** kN-m in magnitude and occurs **1.1** m from the free end.
14. The maximum shear force developed is **2** kN-m in magnitude and occurs **0 to 1.1** from the free end.
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **0**.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **1**.

Scenario 2. A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1 m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.

17. The roller support develops a force of **0.556** kN.
18. The pin support develops a force of **0.444** kN.
19. The bending moment developed at the right end is **0** kN-m.
20. The shear force developed just to the left of the pin is **-0.444** kN.
21. The shear force developed just to the right of the pin is **0.10** kN.
22. **True** or False. There is zero shear force developed at a point 0.9 from the right end
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **1**.
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **2**.

Problem 1



∴ Since the distances between points A to B, B to C, and C to D are all uniform at 2 ft, the vertical heights at points G, F, and E increase by 1 ft each, as the structure forms triangular sections.

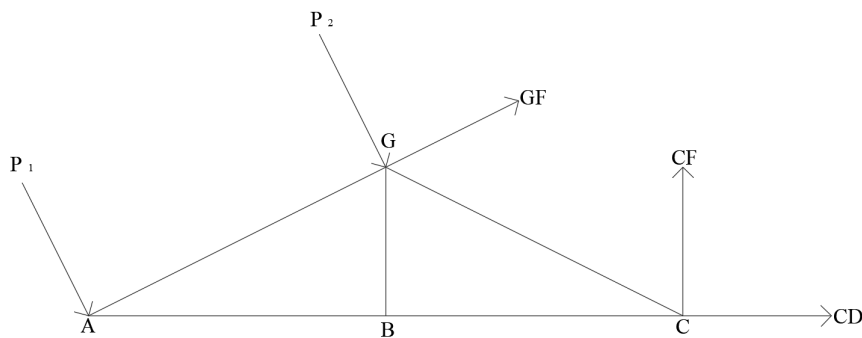
$$\theta_1 = \tan^{-1}\left(\frac{DE}{AD}\right) = \tan^{-1}\left(\frac{3}{6}\right) = 26.57^\circ$$

$$\theta_2 = 90 - 26.57 = 63.43^\circ$$

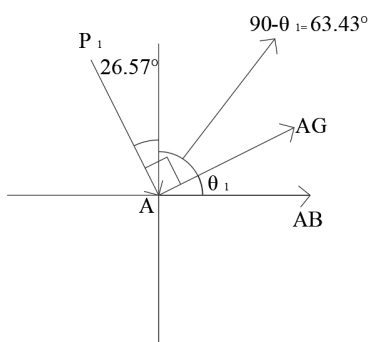
$$\theta_3 = \tan^{-1}\left(\frac{2}{2}\right) = 45^\circ$$

$$BG = (\tan(26.57^\circ))(2) = 1 ; CF = (\tan(26.57^\circ))(4) = 2 ; DE = (\tan(26.57^\circ))(6) = 3$$

Cut to solve FG, CF, and CD



Solve for the angle, x-component and y-component of P forces: (use node A)



Since P_1 , P_2 , P_3 are lying on the same line, they have the same formula in getting their x and y components, but different magnitudes.

X-comp:

Y-comp:

$$P_{1x} = P_1 \sin(26.57^\circ)$$

$$P_{1y} = P_1 \cos(26.57^\circ)$$

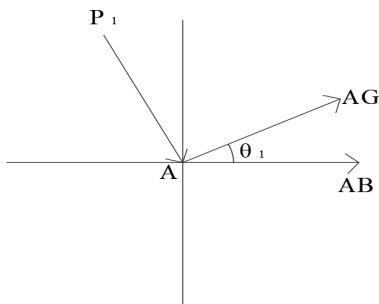
$$P_{2x} = P_2 \sin(26.57^\circ)$$

$$P_{2y} = P_2 \cos(26.57^\circ)$$

$$P_{3x} = P_3 \sin(26.57^\circ)$$

$$P_{3y} = P_3 \cos(26.57^\circ)$$

Node A:



$\uparrow \sum F_y = 0$; Assumption: AG is tension

$$-P_{1y} + AG_y = 0$$

$$-P_1 \cos(26.57^\circ) + AG \sin(26.57^\circ) = 0$$

$$-(5)\cos(26.57^\circ) + AG \sin(26.57^\circ) = 0$$

$$FAG = \left(\frac{5 \cos(26.57^\circ)}{\sin(26.57^\circ)} \right) = 9.998 \text{ kN (T)}$$

$\rightarrow \sum F_x = 0$; Assumption: AB is tension

$$P_{1x} + AG_x + AB_x = 0$$

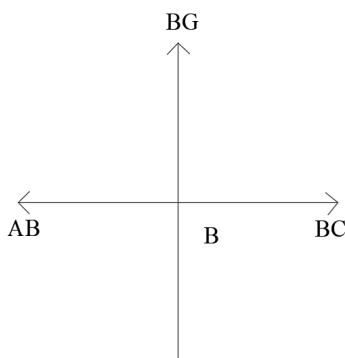
$$P_1 \sin(26.57^\circ) + AG \cos(26.57^\circ) + AB = 0$$

$$AB = -11.179 \text{ kN}$$

$$(5) \sin(26.57^\circ) + (9.998)\cos(26.57^\circ) + AB = 0$$

$$FAB = 11.179 \text{ kN } \odot$$

Node B:



$\uparrow \sum F_y = 0$; Assumption: BG is tension

$$BG_y = 0$$

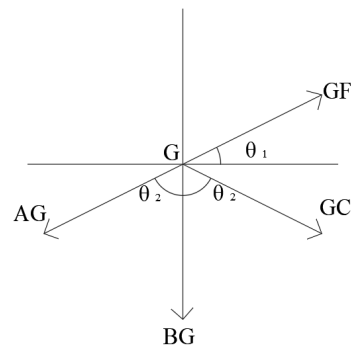
$$BG = 0 \text{ kN (zero-force)}$$

$\rightarrow \sum F_x = 0$; Assumption: BC is tension

$$-AB_x + BC_x = 0$$

$$BC = 11.791 \text{ kN (Tension)}$$

Node G:



$\uparrow \sum F_y = 0$

$$-AG_y - BG_y - GC_y + GF_y = 0$$

$$-AG_y \cos(63.43^\circ) - 0 - GC \cos(63.43^\circ) + GF \sin(26.57^\circ) = 0$$

$$-(9.998 \cos(63.43^\circ)) - GC \cos(63.43^\circ) + GF \sin(26.57^\circ) = 0$$

$$-GC \cos(63.43^\circ) + GF \sin(26.57^\circ) = 4.172 \text{ (eq.1)}$$

$\rightarrow \sum F_x = 0$

$$-AG_x + GC_x + GF_x = 0$$

$$-AG \sin(63.43^\circ) + GC \sin(63.43^\circ) + GF \cos(26.57^\circ) = 0$$

$$-9.998 \sin(63.43^\circ) + GC \sin(63.43^\circ) + GF \cos(26.57^\circ) = 0$$

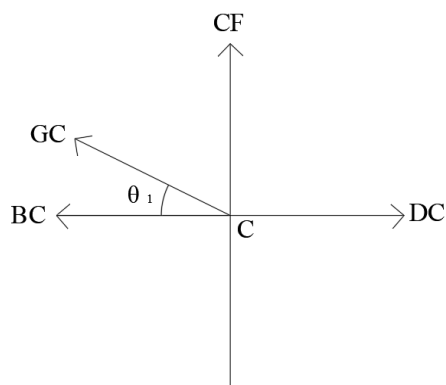
$$GC \sin(63.43^\circ) + GF \cos(26.57^\circ) = 8.992 \text{ (eq.2)}$$

$$\begin{array}{l} \text{eq. 1} \\ \text{eq. 2} \end{array} \begin{bmatrix} \text{GC} & \text{GF} & \text{C} \\ -\cos(63.43) & \sin(26.57) & 4.472 \\ \sin(63.43) & \cos(26.57) & 8.942 \end{bmatrix}$$

$F_{GC} = 0 \text{ kN}$ (zero-force)

$F_{GF} = 9.998 \text{ kN}$ (T)

Node C:



Assumption: CF and DC are in tension

$$\uparrow \Sigma F_y = 0$$

$$G_{Cy} + C_{Fy} = 0$$

$$G_C \sin(26.57^\circ) + C_{Fy} = 0$$

$$(0) \sin(26.57^\circ) + C_{Fy} = 0$$

$$\rightarrow C_{Fy} = 0 \text{ (zero-force)}$$

$$\rightarrow \Sigma F_x = 0$$

$$-B_{Cx} + D_{Cx} = 0$$

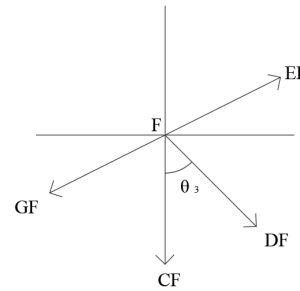
$$-11.179 + D_C = 0$$

$$\rightarrow F_{DC} = 11.179 \text{ kN (T)}$$

Therefore:

- a. • $F_G = 9.998 \text{ kN (T)}$
- $F_{CF} = 0 \text{ kN (zero-force)}$
- $F_{DC} = 11.179 \text{ kN (T)}$

Node F:



Assumption: EF and DF are both in tension

(Figure of force directions)

$$\uparrow \Sigma F_y = 0$$

$$-G_{Fy} - C_{Fy} - D_{Fy} + E_{Fy} = 0$$

$$-G_F \cos(63.43^\circ) - D_F \cos(45^\circ) + E_F \sin(26.57^\circ) = 0$$

$$-9.998 \cos(63.43^\circ) - D_F \cos(45^\circ) + E_F \sin(26.57^\circ) = 0$$

$$-D_F \cos(45^\circ) + E_F \sin(26.57^\circ) = 4.472 \quad (\text{eq1})$$

$$\rightarrow \Sigma F_x = 0$$

$$-G_{Fx} + D_{Fx} + E_{Fx} = 0$$

$$G_F \sin(63.43^\circ) + D_F \sin(45^\circ) + E_F \cos(26.57^\circ) = 0$$

$$-9.998 \sin(63.43^\circ) + D_F \sin(45^\circ) + E_F \cos(26.57^\circ) = 0$$

$$D_F \sin(45^\circ) + E_F \cos(26.57^\circ) = 8.992 \quad (\text{eq2})$$

$$\begin{array}{l} \text{eq. 1} \\ \text{eq. 2} \end{array} \begin{bmatrix} \text{DF} & \text{EF} & \text{C} \\ -\cos(45) & \sin(26.57) & 4.472 \\ \sin(45) & \cos(26.57) & 8.942 \end{bmatrix}$$

Therefore:

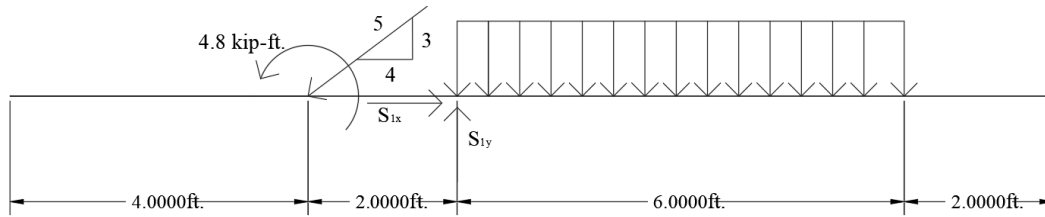
- b. • $F_{DF} = 0 \text{ kN (zero-force)}$
- $F_{EF} = 9.998 \text{ kN}$

C. To maintain stability, I would place the pin at joint A and the roller at joint E or D. This gives a pin with horizontal and vertical reactions and a roller with one vertical reaction, the standard stable simple-support set-up for a planar truss.

D. Based on the forces that we got through our calculations, we got 4 zero-force members, which are FBG, FGC, FCF, and FDF.

E. Based on our calculations, the forces developed in the supports at E is a tension force, and at D is also a tension force

Problem 2



$$M_{5kip} = 5\left(\frac{3}{5}\right)(0.8) + 5\left(\frac{4}{5}\right)(0.6) = 4.8 \text{ kip. ft}$$

$$A. \Sigma M_{s_1} : 5\left(\frac{3}{5}\right)(2) + 4.8 + S_2(8) = 2(6)\left(\frac{6}{2}\right)$$

$$S_2 = \frac{36 - 4.8 - 6}{8}$$

$$S_2 = 3.15 \text{ kip}$$

$$\Sigma F_x : S_{1x} = 5\left(\frac{4}{5}\right) = 4 \text{ kip}$$

$$\Sigma F_y : S_{1y} = 5\left(\frac{4}{5}\right) + 2(6) - 3.15 \text{ kip}$$

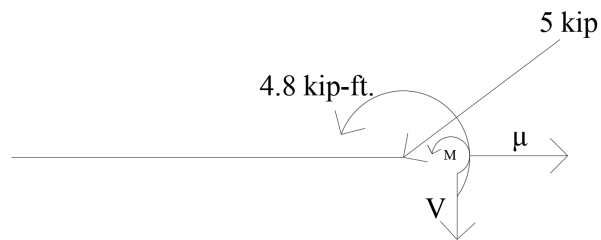
$$S_{1y} = 11.85 \text{ kip}$$

$$B. = 5\left(\frac{4}{5}\right) = 4 \text{ kip}$$

$$C. V = 5\left(\frac{3}{5}\right) = 3 \text{ kip}$$

$$D. 4.8 + \mu = 0$$

$$\mu = 4.8 \text{ kip. ft.}$$



$$\bullet \quad 0 \leq x < 4$$

$$F_N = 0$$

$$V = 0$$

$$\mu = 0$$



$$\bullet \quad 4 \leq x < 6$$

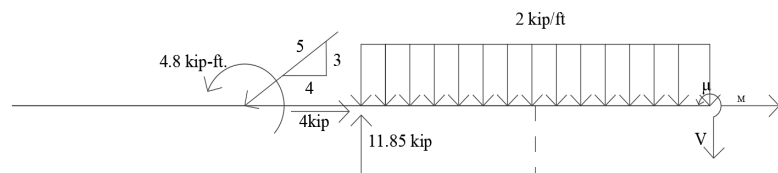
$$F_N = 4 \text{ kip}$$

$$V = -3 \text{ kip}$$

$$\mu = -4.8 - 5\left(\frac{3}{5}\right)(x - 4)$$

$$= -4.8 - 3x + 12$$

$$= 7.2 - 3x$$



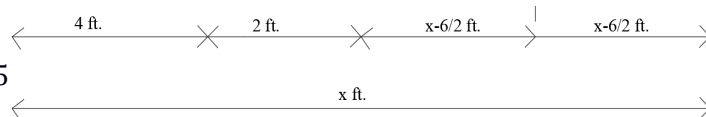
$$\bullet \quad 6 \leq x < 12$$

$$F_N = 0$$

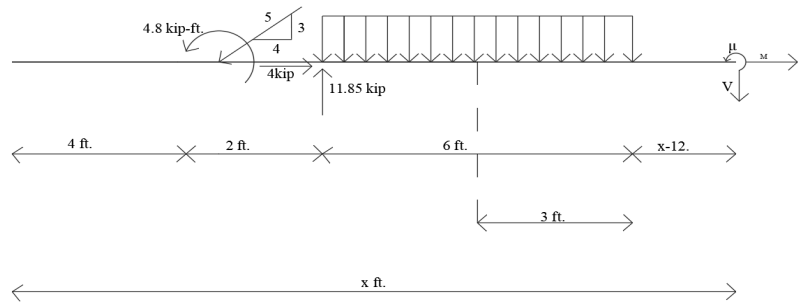
$$V = 5\left(\frac{3}{5}\right) - 2(x - 6) + 11.85$$

$$= 20.85 - 2x$$

$$\mu = -4.8 - 5\left(\frac{3}{5}\right)(x - 4) - 2(x - 6)\left(\frac{x-6}{2}\right) + 11.85(x - 6)$$



$$\begin{aligned}
 &= -4.8 - 3x + 12 - (x^2 - 12x + 36) + 11.85x - 71.1 \\
 &= -99.9 + 20.85x - x^2 \\
 &= -x^2 + 20.85x - 99.9
 \end{aligned}$$



• $12 \leq x \leq 14$
 $F_N = 0 \text{ kip}$

$$\begin{aligned}
 V &= -5\left(\frac{3}{5}\right) - 2(6) + 11.85 \\
 &= -3.15 \text{ kip}
 \end{aligned}$$

$$\begin{aligned}
 \mu &= -4.8 - 5\left(\frac{3}{5}\right)(x - 4) - 2(6)(x - 9) + 11.85(x - 6) \\
 &= -4.8 - 3x + 12 - 12x + 108 + 11.85x - 71.1 \\
 &= 44.1 - 3.15x
 \end{aligned}$$

E.

$$V = \begin{cases} 0 & 0 \leq x \leq 4 \\ -3 & 4 \leq x \leq 12 \\ 20.85 - 2x & 6 \leq x \leq 12 \\ -3.15 & 12 \leq x \leq 14 \end{cases}$$

F.

$$\mu = \begin{cases} 0 & 0 \leq x \leq 4 \\ 7.2 - 3x & 4 \leq x \leq 12 \\ -x + 20.85x - 99.9 & 6 \leq x \leq 12 \\ 44.1 - 3.15x & 12 \leq x \leq 14 \end{cases}$$